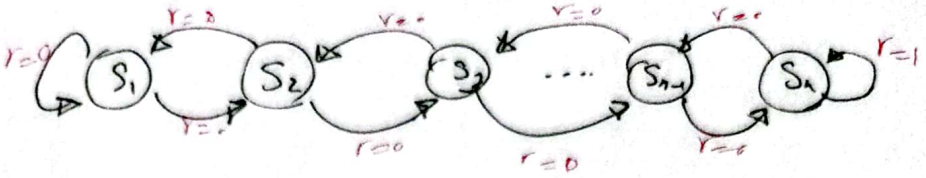
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RL
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Problem 1:



a_1 ← , a_0 →

$\mathcal{S} = \{a_0, a_1\}$

(A)

since everything is deterministic $p(s'|s,a) = \delta(s' - s_{next})$

writing Bellman's optimality equation for $V^*(s)$:

$$\begin{cases} V^*(s_n) = \max_a \{ R(s,a) + \gamma \cdot \sum_{s' \in \mathcal{S}} \delta(s_n - s') V^*(s') \} \\ V^*(s_n) = 1 + \gamma \cdot V^*(s_n) \Rightarrow \boxed{V^*(s_n) = \frac{1}{1-\gamma}} = V^*(G) \end{cases} \quad (*)$$

$R(s_n, a_0) = 1$, $a^* = a_0$, $s' = s_n$

$$\begin{cases} \forall 1 \le i < n: \ V^*(s_i) = \max_a \{ R(s,a) + \gamma \cdot \sum_{s' \in \mathcal{S}} \delta(s_{i+1} - s') V^*(s') \} \\ \boxed{V^*(s_i) = 0 + \gamma \cdot V^*(s_{i+1})} \ (**) \end{cases} \longrightarrow V^*(s_{n-k}) = \gamma^k V^*(s_n)$$

$(*), (**) \xrightarrow{1 \le i \le n}$ $\boxed{V^*(s_i) = \frac{\gamma^{n-i}}{1-\gamma}}$

(B) if $\gamma \in [0,1)$, the optimal policy remains unchanged even if we change the value of γ. → the optimal policy will always be to get to the terminal state G (which is independent of the choice of γ) cause it's chosen in a greedy way.

(C) So we write the Bellman's optimality equation again:

$$V^*(s_n) = \max_a \{ R(s,a) + \gamma \cdot \sum_{s' \in \mathcal{S}} V^*(s') \} = 1 + c + \gamma \cdot V^*(s_n) \Rightarrow V^*(s_n) = \frac{1+c}{1-\gamma}$$

$$\forall 0 \le i < n: \ V^*(s_i) = (0 + c) + \gamma \cdot V^*(s_{i+1}) \quad \text{let } A = \frac{1+c}{1-\gamma}, \ B = c$$

$$\Rightarrow \begin{cases} X_n = A \\ X_i = B + \gamma X_{i+1} \end{cases} \longrightarrow X_{n-k} = \gamma^k A + B \sum_{j=0}^{k-1} \gamma^j = \gamma^k A + B \frac{1-\gamma^k}{1-\gamma}$$

thus: $V^*(s_i) = \gamma^{n-i} \left(\frac{1+c}{1-\gamma} \right) + c \frac{1-\gamma^{n-i}}{1-\gamma} = \frac{\gamma^{n-i} + c}{1-\gamma} \Rightarrow \begin{cases} V^*(s_n) = (1+c)/(1-\gamma) \\ V^*(s_i) = \frac{\gamma^{n-i} + c}{1-\gamma} \end{cases}$

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NO! adding a constant (or multiplying the reward by a constant) will not change the optimal policy [it may change the convergence rate of [Policy iteration / Value iteration] algorithms], but it doesn't change the optimal policy. for any $\gamma \in [0,1)$, still the optimal action is to go to the right to get to terminal State 6.

$$V^*(S_i) = \frac{\gamma^{n-i}}{1-\gamma} \longleftrightarrow V^*_{new}(S_i) = \frac{\gamma^{n-i} + c}{1-\gamma}$$

(D) (I) So we saw that adding a fixed value d to the reward functions, results in $V^*(S_i) = \frac{\gamma^{n-i} + d}{1-\gamma}$.

(II) Now, in the original setting, scaling the reward function by β results in $\Rightarrow V^*(S_1) = \frac{\beta}{1-\gamma} \Rightarrow$ from $V^*(S_i) = \gamma V^*(S_{i+1})$ we get:

$$V^*(S_i) = \frac{\beta \gamma^{n-i}}{1-\gamma} \quad \forall\, 1 \le i \le n$$

Now letting $r^{new} = a(c + r_{old}) = ac + a \cdot r_{old}$

$\beta = a$ (i) first Scaling $\rightarrow V^{new,*}(S_i) = \frac{a\gamma^{n-i}}{1-\gamma}$

$\alpha = ac$ (ii) now shifting it $\rightarrow$ $V^{new,*}(S_i) = \frac{a\gamma^{n-i} + ac}{1-\gamma} = a \cdot \frac{\gamma^{n-i} + c}{1-\gamma}$

easier explanation

$$V^{\pi^*}_{new}(S) = \sum_{t=0}^{60} \gamma^t \left[a R_{old}(S_t, a_t) + ac \right] = a V^{\pi^*}_{old}(S) + \frac{ac}{1-\gamma}$$

whether this changes the optimal policy $\pi^* = \arg\max_{\pi} V^{\pi}(S)$ depends on the sign of 'a':

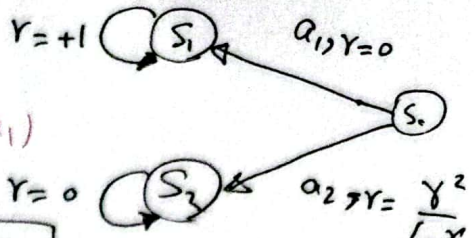
if $a > 0 \Rightarrow$ it preserves the relative ordering $V^{\pi_1}_{old}(S) \gtrless V^{\pi_2}_{old}(S)$

$\Rightarrow V^{\pi_1}_{new} \gtrless V^{\pi_2}_{new}(S)$ [$a > 0$]

However if $a < 0 \Rightarrow V^{\pi_1}_{new}(S) \lesseqgtr V^{\pi_2}_{new}(S)$

↳ it turns maximisation to minimisation, the agent will now prefer policies that yield the lowest original returns. Thus the optimal policy changes completely if $a < 0$.

Problem 2



(A) The total discounted return after taking action a_1 from state S_0 is actually $Q(S_0, a_1)$

$$\rightarrow Q(S_0,a_1) = \sum_{t=0}^{\infty} \gamma^t \cdot R_t = \cancel{R_0} + \sum_{t=1}^{\infty} \gamma^t \cdot 1 = \boxed{\frac{\gamma}{1-\gamma}}$$

(B) $$Q(S_0,a_2) = \sum_{t=0}^{\infty} \gamma^t \cdot R_t = R_0 + \sum_{t=0}^{\infty} \gamma^t \cdot 0 = R_0 = \boxed{\frac{\gamma^2}{1-\gamma}}$$

Now since $\begin{cases} |\gamma| < 1 \\ \gamma > 0 \end{cases} \Rightarrow Q(S_0,a_1) > Q(S_0,a_2)$ & since the optimal action is: $a^* = \max_a Q(a,S) \rightarrow \boxed{a^* = a_1}$ is the optimal action

(C) $V^n(S)\big|_{n=0} = 0 \quad \forall S \in \mathcal{S} \Rightarrow$ we take the optimal decision if $\Rightarrow Q^n(S_0,a_1) \geq Q^n(S_0,a_2)$

$$\begin{cases} \text{for } S_1: \quad V^n(S_1) = 1 + \gamma \cdot V^{n-1}(S_1) = \sum_{k=0}^{n-1} \gamma^k = \frac{1-\gamma^n}{1-\gamma} \\ V^0(S_1) = 0 \end{cases}$$

$$\begin{cases} \text{for } S_2: \quad V^n(S_2) = 0 + \gamma \cdot V^{n-1}(S_2) = 0 \quad \forall n \\ V^0(S_2) = 0 \end{cases}$$

$\Rightarrow$ so lets calculate $Q^n(\cdot,\cdot)$:

$$\begin{cases} Q^n(S_0,a_1) = R(S_0,a_1) + \gamma \cdot V^n(S_1) = 0 + \gamma \cdot \frac{1-\gamma^n}{1-\gamma} = \frac{\gamma(1-\gamma^n)}{1-\gamma} \\ Q^n(S_0,a_2) = R(S_0,a_2) + \gamma \cdot V^n(S_2) = \frac{\gamma^2}{1-\gamma} + 0 = \frac{\gamma^2}{1-\gamma} \end{cases}$$

$$Q^n(S_0,a_1) > Q^n(S_0,a_2) \iff \frac{\gamma(1-\gamma^n)}{(1-\gamma)} \geq \frac{\gamma^2}{1-\gamma}$$

$$\iff 1-\gamma^n \geq \gamma \quad \Leftarrow \quad \boxed{n^* \geq \frac{\log(1-\gamma)}{\log \gamma}} \quad \text{Q.E.D.}$$

$\iff$ for $0 < \gamma < 1$

